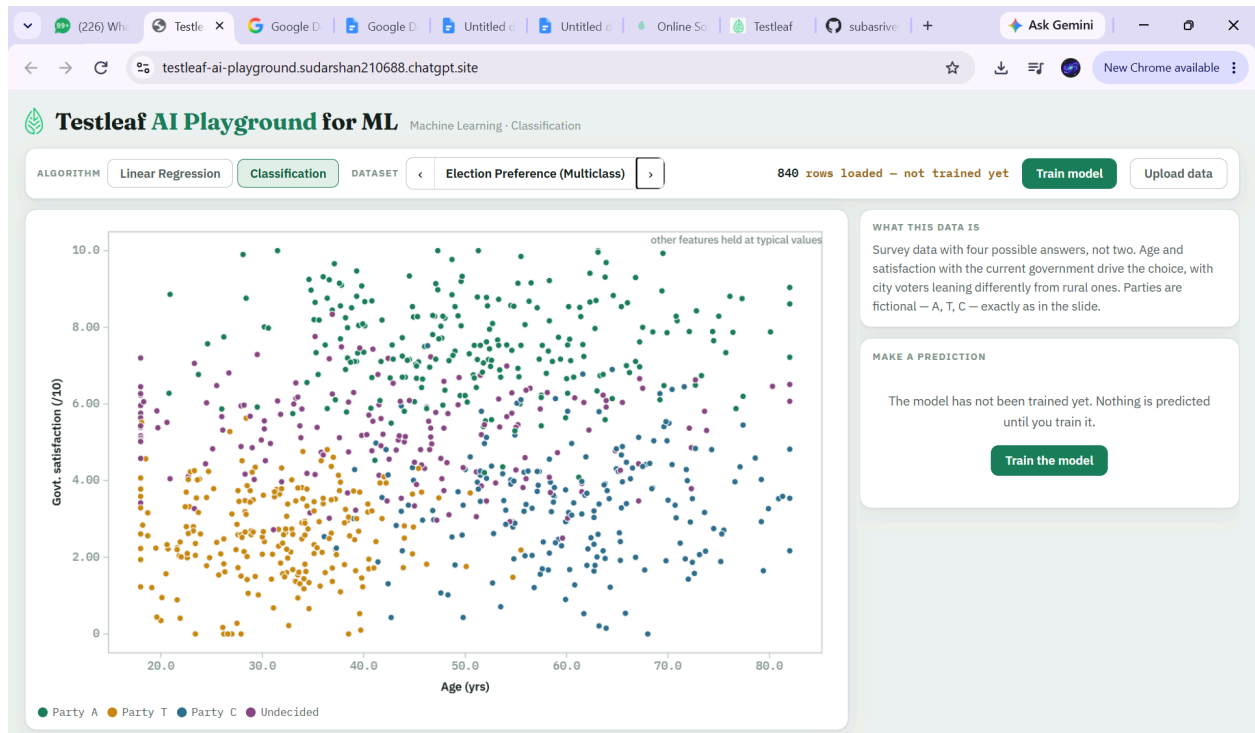
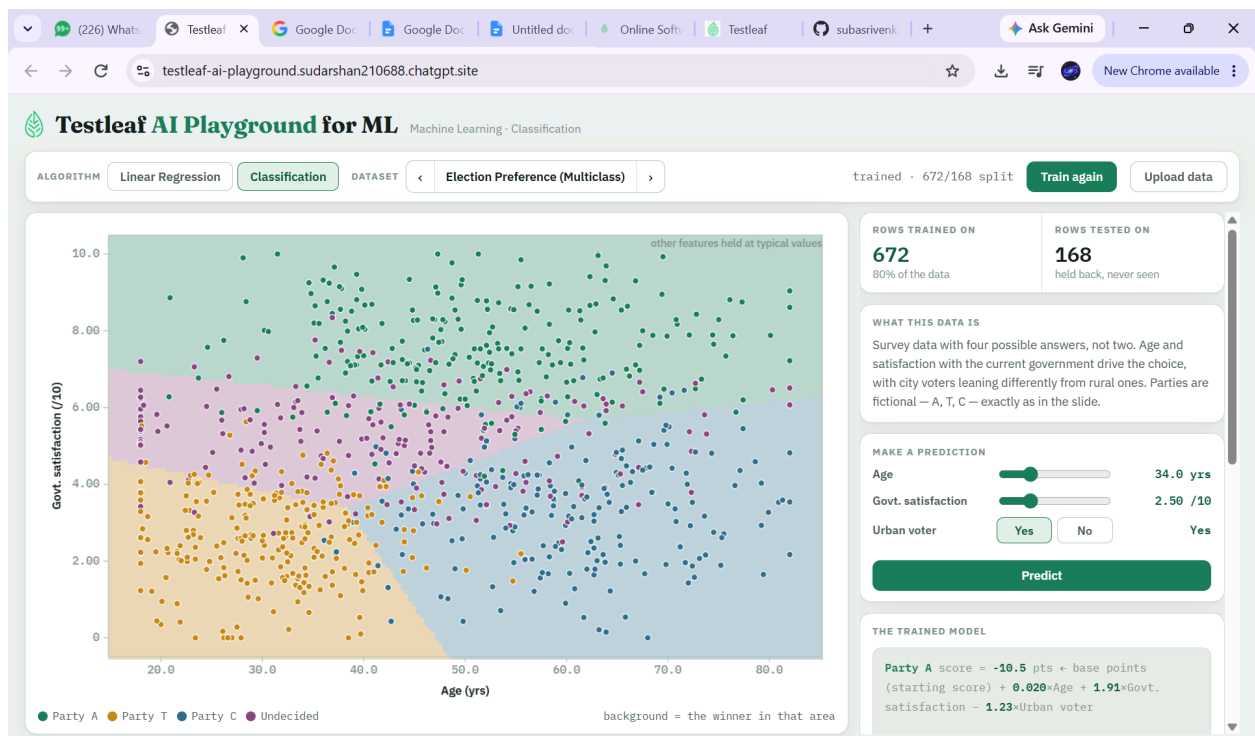


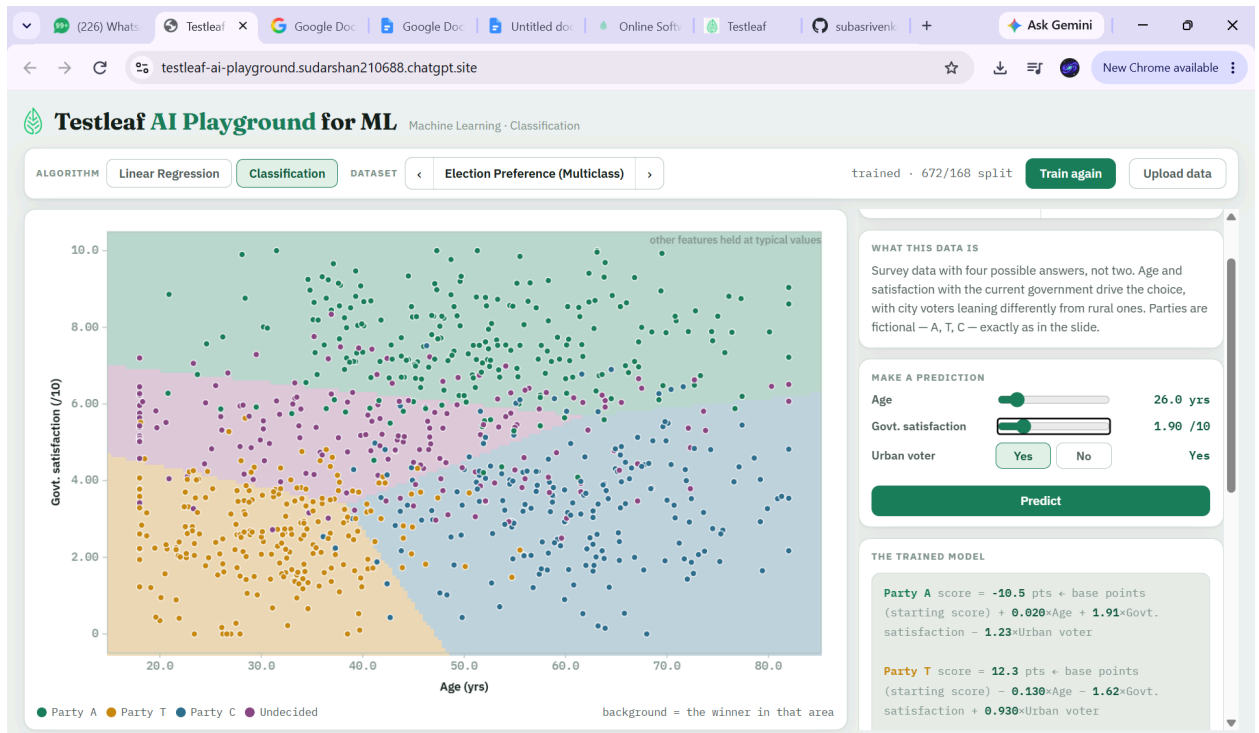


## CLASSIFICATION ALGORITHM:

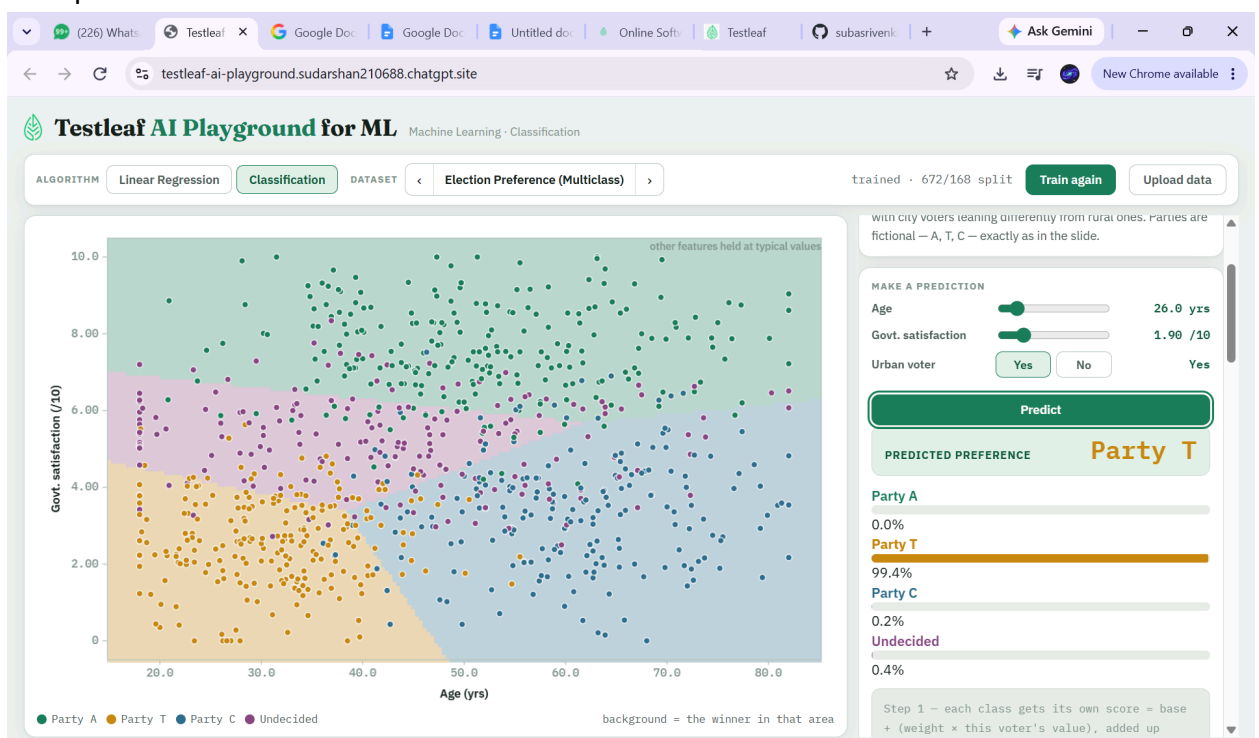
Click Train model:



a) Set age <30 and govt satisfaction less, urban:



Click predict:



Step 1 – each class gets its own score = base + (weight × this voter's value), added up

Party A

```

-10.5 pts ← base points
+ 0.020 × 26.0 (Age) = +0.528
+ 1.91 × 1.90 (Govt. satisfaction) = +3.64
- 1.23 × Yes (Urban voter) = -1.23

```

---

**= -7.52 pts**

#### Party T

```

12.3 pts ← base points
- 0.130 × 26.0 (Age) = -3.39
- 1.62 × 1.90 (Govt. satisfaction) = -3.09
+ 0.930 × Yes (Urban voter) = +0.930

```

---

**= 6.73 pts**

#### Party C

```

1.58 pts ← base points
+ 0.097 × 26.0 (Age) = +2.51
- 1.04 × 1.90 (Govt. satisfaction) = -1.97
- 1.50 × Yes (Urban voter) = -1.50

```

---

**= 0.622 pts**

#### Undecided

```

2.10 pts ← base points
- 0.028 × 26.0 (Age) = -0.716
+ 0.225 × 1.90 (Govt. satisfaction) = +0.427
- 0.554 × Yes (Urban voter) = -0.554

```

---

**= 1.25 pts**

### highest score wins → Party T

Step 2a – convert each score to  $e^{\text{score}}$  (always positive)

```

Party A     $e^{-7.52} = 0.001$ 
Party T     $e^{6.73} = 835$ 
Party C     $e^{0.622} = 1.86$ 
Undecided   $e^{1.25} = 3.50$ 

```

---

### Total (add all 4 up) = 840

Step 2b – each class's share = its own  $e^{\text{score}}$  ÷ that same total (840)

```

Party A     $0.001 \div 840 = 0.0\%$ 
Party T     $835 \div 840 = 99.4\%$ 
Party C     $1.86 \div 840 = 0.2\%$ 
Undecided   $3.50 \div 840 = 0.4\%$ 

```

---

All 4 shares add back up to 100% – that's the whole point of dividing by the same total.  $e$  ( $\approx 2.71828$ ) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

### The trained model

**Party A** score = -10.5 pts ← base points (starting score) + 0.020×Age + 1.91×Govt. satisfaction - 1.23×Urban voter

**Party T** score = 12.3 pts ← base points (starting score) - 0.130×Age - 1.62×Govt. satisfaction + 0.930×Urban voter

**Party C** score = 1.58 pts ← base points (starting score) + 0.097×Age - 1.04×Govt. satisfaction - 1.50×Urban voter

**Undecided** score = 2.10 pts ← base points (starting score) - 0.028×Age + 0.225×Govt. satisfaction - 0.554×Urban voter

**Sample calculation — one real row**

45.7 yrs · 6.43 /10 · not urban voter

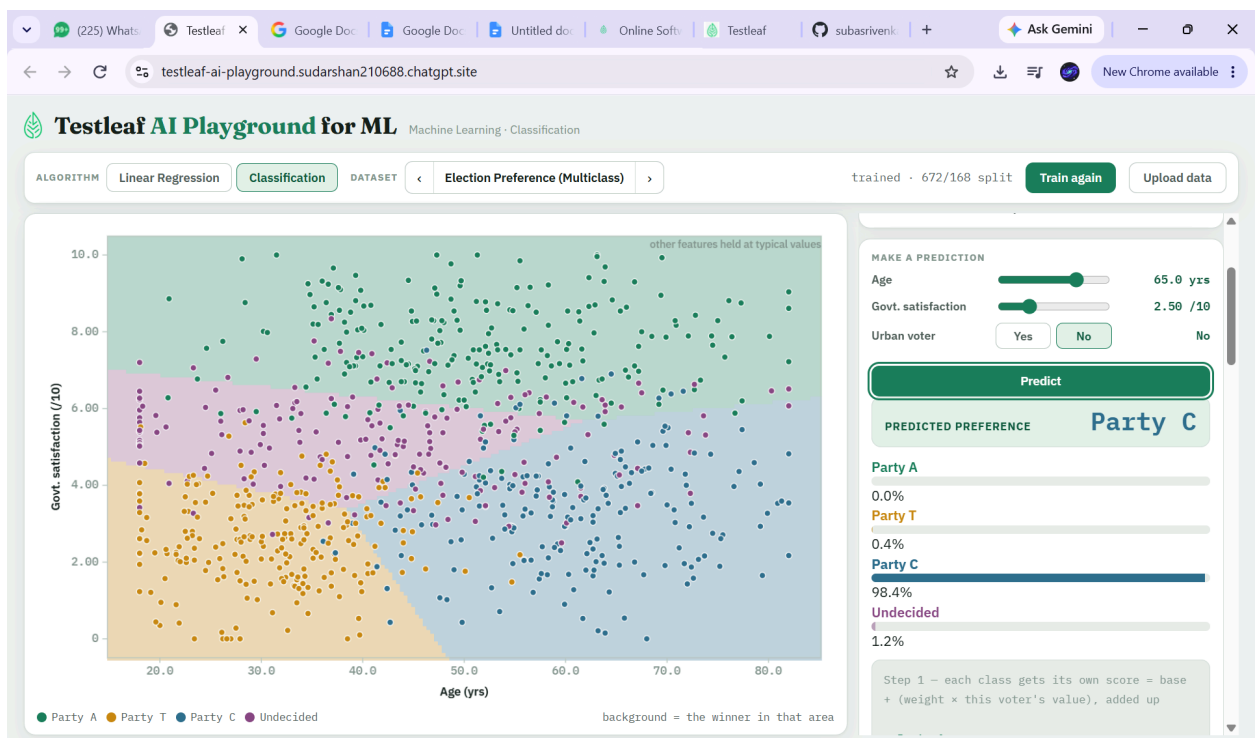
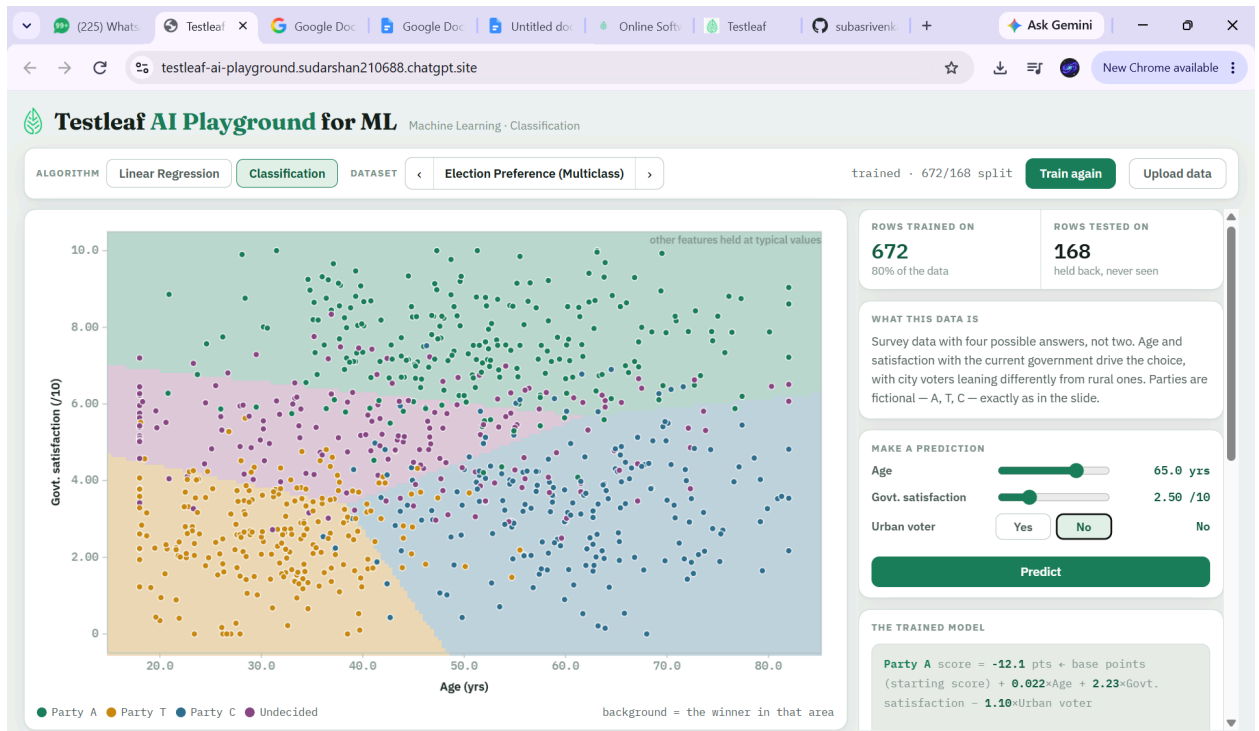
Party A    2.78  
Party T    -4.13  
Party C    -0.662  
Undecided 2.28

---

**highest wins → Party A**

actual            Party A

**b) Set age >50 and govt satisfaction less, non urban:**



Step 1 — each class gets its own score = base + (weight × this voter's value), added up

#### Party A

-12.1 pts ← base points  
 + 0.022 × 65.0 (Age) = +1.46  
 + 2.23 × 2.50 (Govt. satisfaction) = +5.58

$$- 1.10 \times \text{No (Urban voter)} = +0$$

---


$$= -5.07 \text{ pts}$$

#### Party T

$$\begin{aligned} 10.6 \text{ pts} &\leftarrow \text{base points} \\ - 0.128 \times 65.0 \text{ (Age)} &= -8.35 \\ - 1.31 \times 2.50 \text{ (Govt. satisfaction)} &= -3.27 \\ + 1.05 \times \text{No (Urban voter)} &= +0 \end{aligned}$$

---


$$= -0.976 \text{ pts}$$

#### Party C

$$\begin{aligned} -0.057 \text{ pts} &\leftarrow \text{base points} \\ + 0.099 \times 65.0 \text{ (Age)} &= +6.42 \\ - 0.720 \times 2.50 \text{ (Govt. satisfaction)} &= -1.80 \\ - 1.38 \times \text{No (Urban voter)} &= +0 \end{aligned}$$

---


$$= 4.56 \text{ pts}$$

#### Undecided

$$\begin{aligned} 0.459 \text{ pts} &\leftarrow \text{base points} \\ - 0.025 \times 65.0 \text{ (Age)} &= -1.65 \\ + 0.541 \times 2.50 \text{ (Govt. satisfaction)} &= +1.35 \\ - 0.432 \times \text{No (Urban voter)} &= +0 \end{aligned}$$

---


$$= 0.158 \text{ pts}$$

### highest score wins → Party C

Step 2a – convert each score to e<sup>score</sup> (always positive)

$$\begin{aligned} \text{Party A} \quad e^{-5.07} &= 0.006 \\ \text{Party T} \quad e^{-0.976} &= 0.377 \\ \text{Party C} \quad e^{4.56} &= 96.0 \\ \text{Undecided} \quad e^{0.158} &= 1.17 \end{aligned}$$

---


$$\text{Total (add all 4 up)} = 97.6$$

Step 2b – each class's share = its own e<sup>score</sup> ÷ that same total (97.6)

$$\begin{aligned} \text{Party A} \quad 0.006 \div 97.6 &= 0.0\% \\ \text{Party T} \quad 0.377 \div 97.6 &= 0.4\% \\ \text{Party C} \quad 96.0 \div 97.6 &= 98.4\% \\ \text{Undecided} \quad 1.17 \div 97.6 &= 1.2\% \end{aligned}$$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

### The trained model

**Party A** score = -12.1 pts ← base points (starting score) + 0.022×Age + 2.23×Govt. satisfaction - 1.10×Urban voter

**Party T** score = 10.6 pts ← base points (starting score) - 0.128×Age - 1.31×Govt. satisfaction + 1.05×Urban voter

**Party C** score =  $-0.057 \text{ pts} \leftarrow \text{base points (starting score)} + 0.099 \times \text{Age} - 0.720 \times \text{Govt. satisfaction} - 1.38 \times \text{Urban voter}$

**Undecided** score =  $0.459 \text{ pts} \leftarrow \text{base points (starting score)} - 0.025 \times \text{Age} + 0.541 \times \text{Govt. satisfaction} - 0.432 \times \text{Urban voter}$

#### Sample calculation — one real row

45.7 yrs · 6.43 /10 · not urban voter

Party A 3.27  
Party T -3.64  
Party C -0.170  
Undecided 2.78

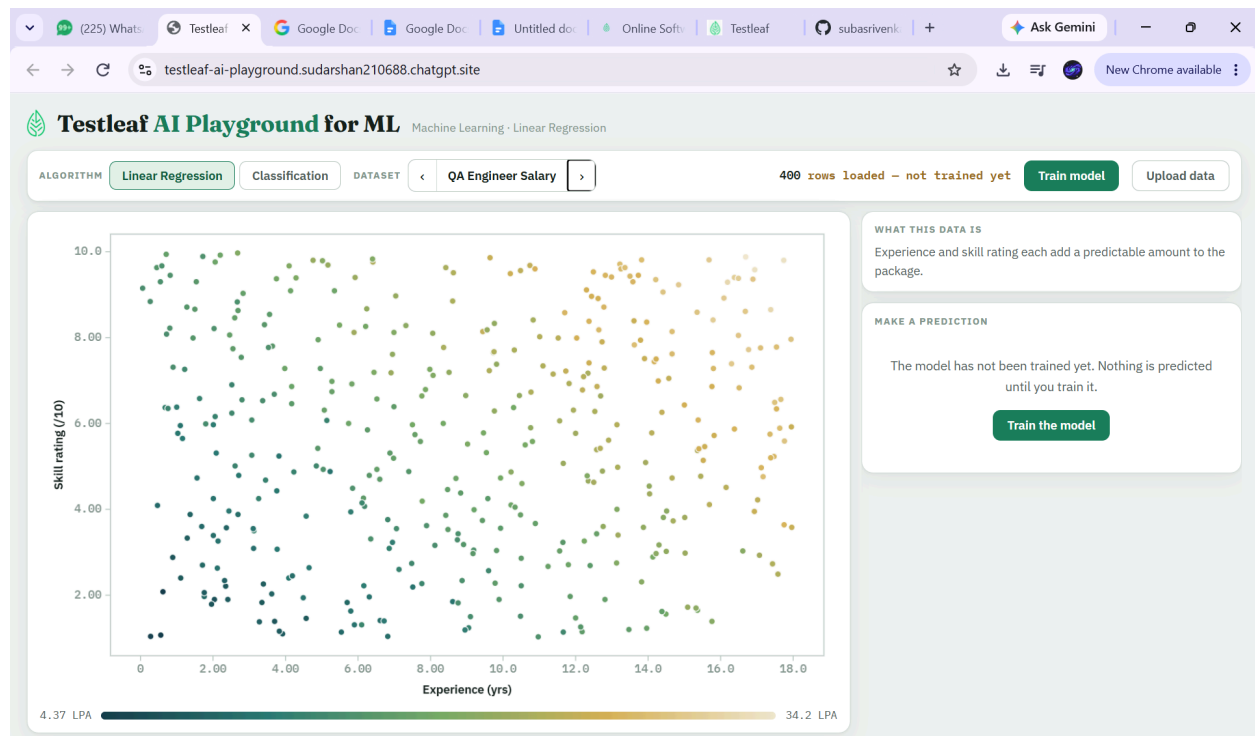
**highest wins → Party A**

actual Party A

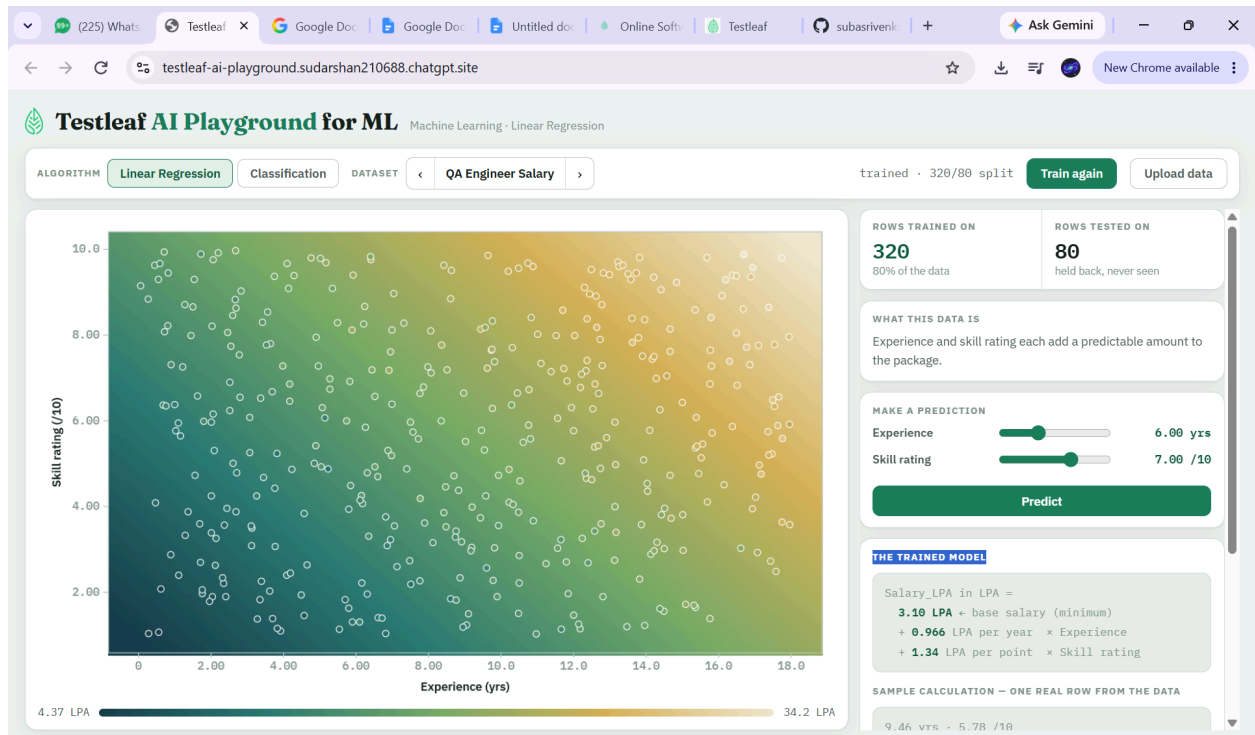
With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

## LINEAR REGRESSION:

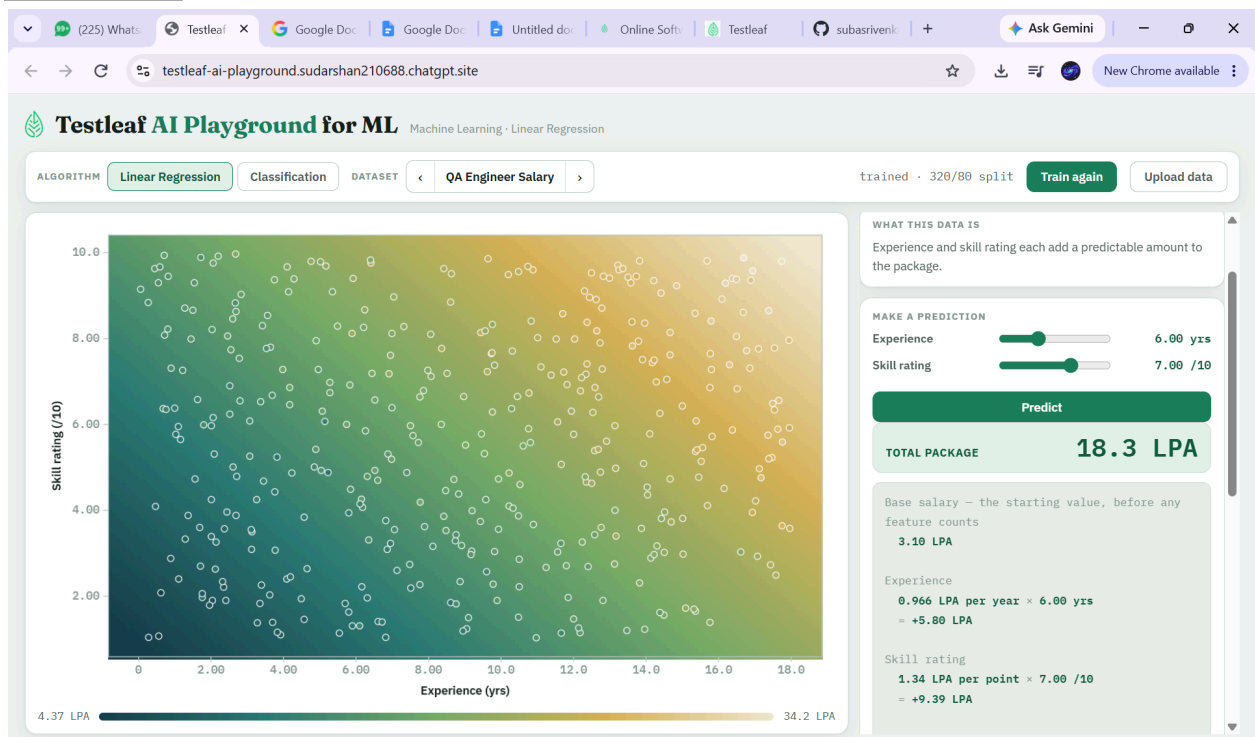
### a) QA Salary:



**Click train model:**



## Click Predict



Total package **18.3 LPA**

Base salary — the starting value, before any feature counts  
3.10 LPA

Experience  
 $0.966 \text{ LPA per year} \times 6.00 \text{ yrs}$   
 $= +5.80 \text{ LPA}$

Skill rating  
 $1.34 \text{ LPA per point} \times 7.00 / 10$   
 $= +9.39 \text{ LPA}$

---

**TOTAL     18.3 LPA**

**The trained model**

Salary\_LPA in LPA =  
3.10 LPA ← base salary (minimum)  
+ 0.966 LPA per year × Experience  
+ 1.34 LPA per point × Skill rating

**Sample calculation — one real row from the data**

9.46 yrs · 5.78 /10

3.10 (base)  
+ 0.966 × 9.46 = +9.14  
+ 1.34 × 5.78 = +7.75

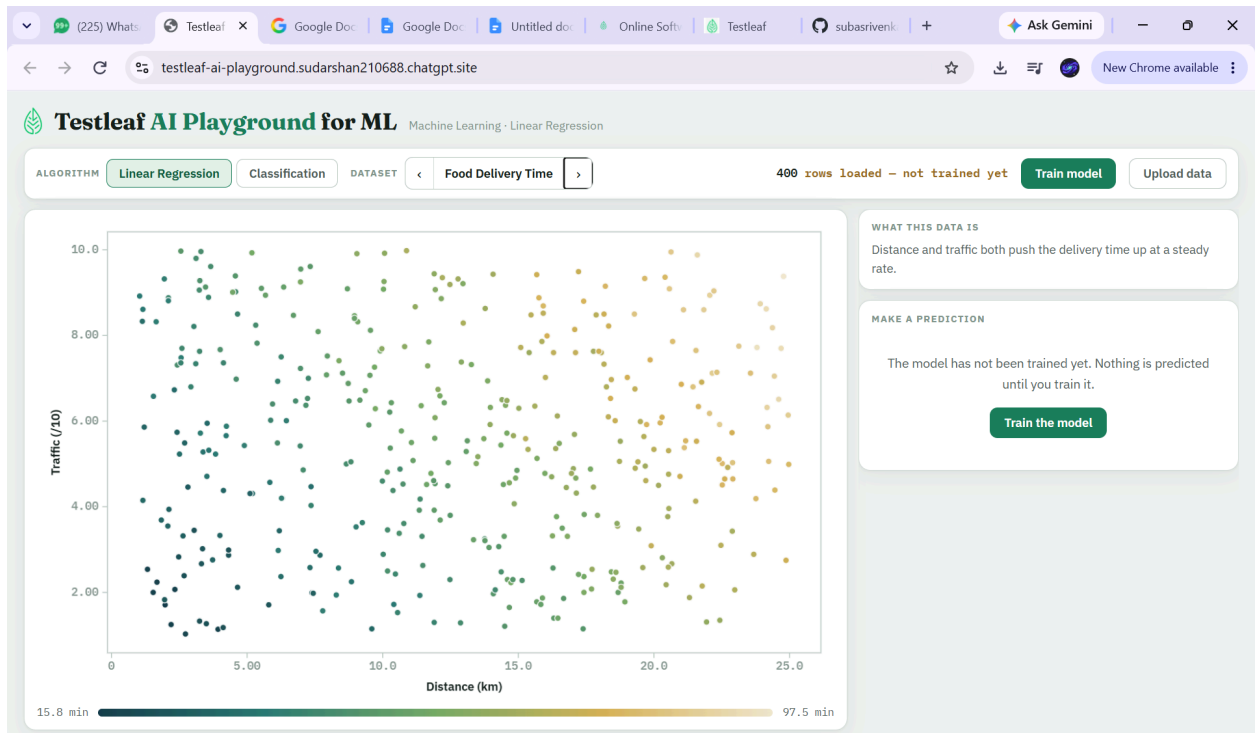
---

**model says     20.0 LPA**

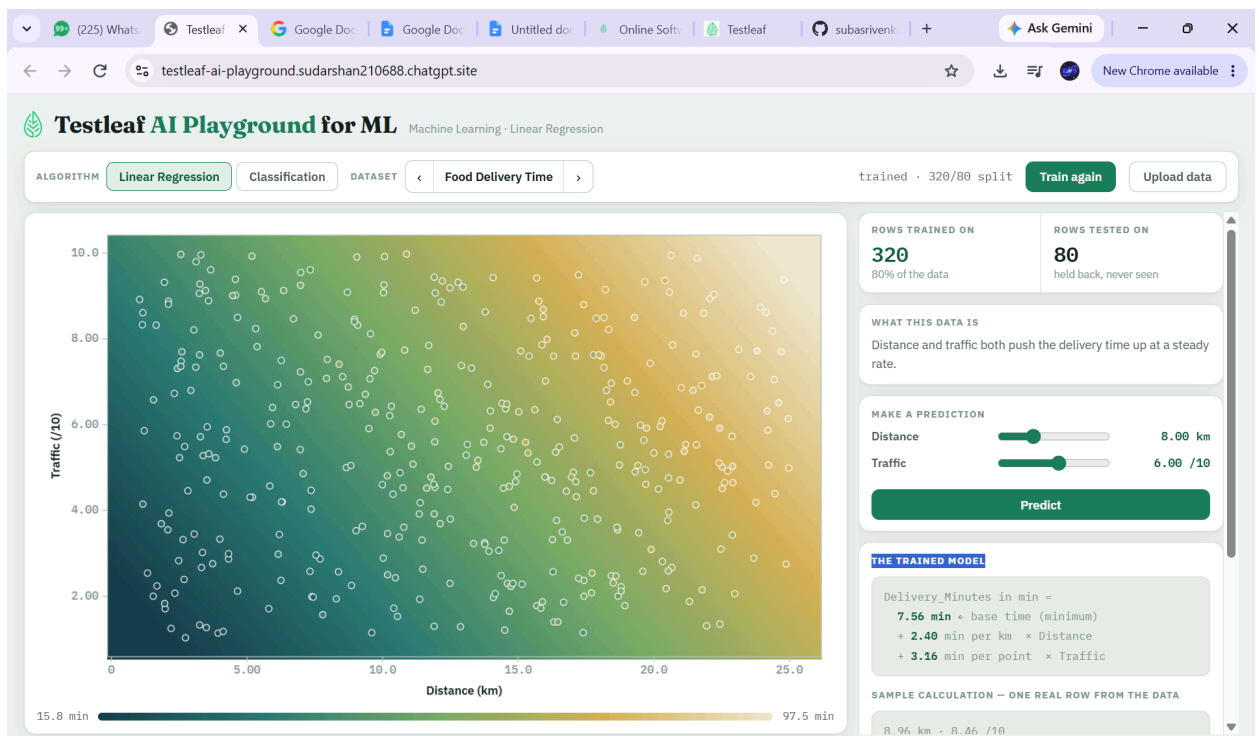
actual value   19.4 LPA

Read it as a rate card. The first number is the **base salary** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

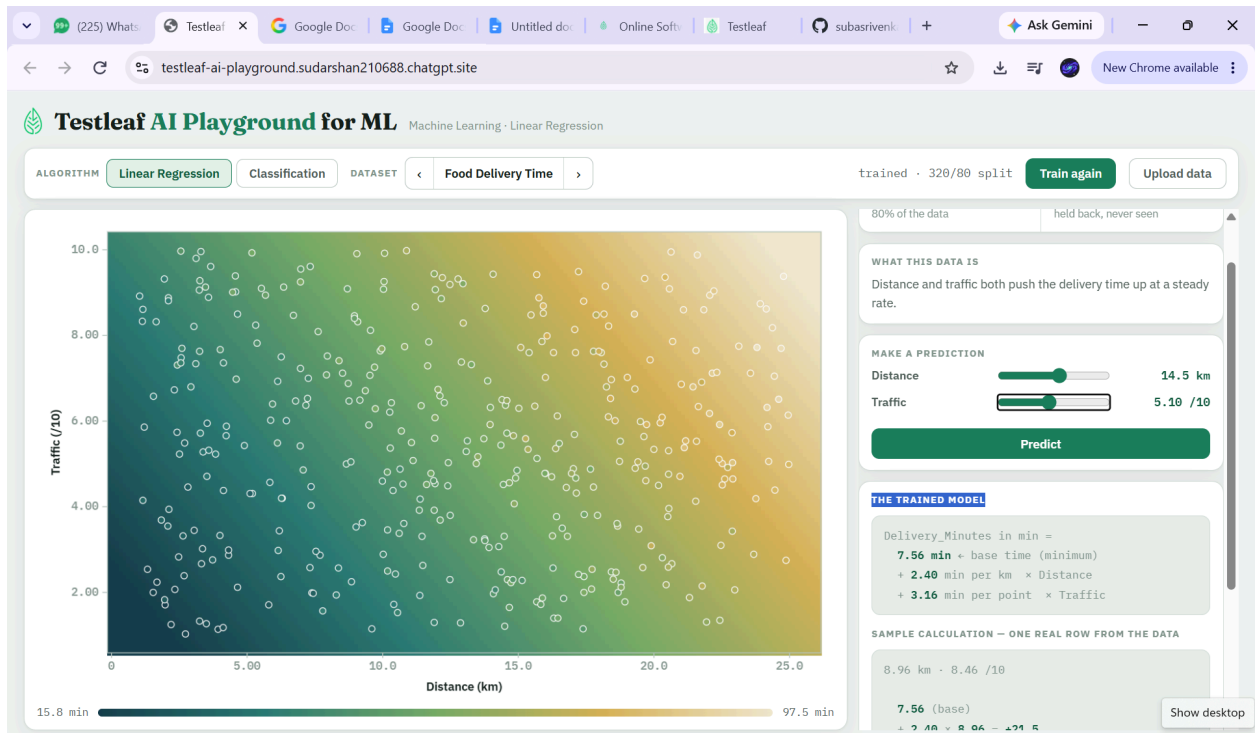
**b) FOOD DELIVERY TIME:**



**Click Train model:**



**Set the parameters for prediction:**



**Click predict:**

**Total time 58.5 min**

Base time — the starting value, before any feature counts  
7.56 min

Distance  
2.40 min per km × 14.5 km  
= +34.8 min

Traffic  
3.16 min per point × 5.10 /10  
= +16.1 min

**TOTAL 58.5 min**

**The trained model**

Delivery\_Minutes in min =  
7.56 min + base time (minimum)  
+ 2.40 min per km × Distance

+ 3.16 min per point × Traffic

**Sample calculation — one real row from the data**

8.96 km · 8.46 /10

7.56 (base)  
+ 2.40 × 8.96 = +21.5  
+ 3.16 × 8.46 = +26.7

---

**model says    55.8 min**

actual value   55.4 min

Read it as a rate card. The first number is the **base time** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.